

Tran Khanh Thanh

Robot Learning and Research Engineering

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📍 Hanoi, Vietnam

🔗 khanhthanhdev.github.io

🐙 [khanhthanhdev](https://github.com/khanhthanhdev)

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Education

Bachelor of Science - **VinUniversity**, Electrical and Computer Engineering
 • Research objective: develop adaptive assistive robots through RL, imitation learning, and rigorous physical-system validation; pursue a PhD in robotics or embodied AI.
 Hanoi, Vietnam
 2025 – 2029

Experience

LeRobot / Hiwonder NexArm fork, Robot-learning contributor
 • Contributed MuJoCo simulation support for a six-action NexArm contract, with visual meshes, stable primitive collisions, front and wrist cameras, and seeded pick-and-place resets.
 • Built scripted-demonstration generation, dataset provenance, ACT-compatible rollout and benchmark tooling, Rerun integration, and focused validation tests for action mapping, contacts, cameras, and gripper behavior.
 Open-source
 2026 – present
 1 year

STEAM for Vietnam, Software Engineer
 • Develop software for robotics competition operations, including real-time multi-user workflows, local-network operation, scheduling, ranking, and rules processing.
 Hanoi, Vietnam
 Apr 2025 – present
 1 year 5 months

PRISM research collaboration, Research contributor
 • Co-authored PRISM, a benchmark for AI peer reviewers, and implemented an evidence-grounded validation workflow covering dataset contracts, blinded annotations, judge caching, statistical reporting, and offline synthetic tests.
 Remote
 Sept 2025 – present
 1 year

Selected Engineering Project

Robotics Tournament Management System
 Cross-platform desktop and local-network system for live robotics competition operations.
 • Built with TypeScript, Bun, Electrobun, React, Hono, SQLite, Server-Sent Events, and TinyBase for concurrent venue workflows without an internet dependency.

Skills

Robot learning: Python, PyTorch, LeRobot, imitation learning, learning from demonstration, ACT-compatible training and rollout

Simulation and validation: MuJoCo, teleoperation, synthetic datasets, Rerun, software testing, debugging, reproducible evaluation

Engineering: C++, TypeScript, Git, Linux, Docker, technical English

Awards

First Prize, Vietnam AI Contest 2024
 VLAB Innovation and Boston Global Forum
 Dec 2024

Gold Medal, FIRST Global Challenge 2023 Finalist Winning Alliance
 FIRST Global
 Oct 2023